

“Impact of Oil Price Shocks on Stock Market Volatility in Pakistan: Evidence from Non-Linear ARDL Technique”



*A Dissertation submitted to the Pakistan Institute of Development Economics for the
Partial Fulfilment of the Requirement for the Degree of Master of Philosophy in
Economics and Finance*

(2019)

Submitted by:

Kaifayat Ullah

Registration No.

PIDE2016FMPHILEAF05

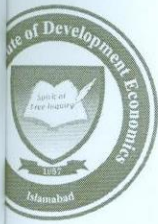
Supervised by:

Dr. Saud Ahmad Khan

Department of Business Studies

PAKISTAN INSTITUTE OF DEVELOPMENT ECONOMICS

ISLAMABAD



PAKISTAN INSTITUTE OF DEVELOPMENT ECONOMICS, ISLAMABAD

CERTIFICATE

This is to certify that this thesis entitled **“Impact of oil price shocks on stock market volatility in Pakistan- Evidence from Non-Linear ARDL Technique.”** submitted by **Mr. Kaifayat Ullah** is accepted in its present form by the Department of Business Studies, MPhil Economics and Finance, Pakistan Institute of Development Economics (PIDE) Islamabad as satisfying the requirements for partial fulfillment of the Degree of Master of Philosophy in Economics and Finance.

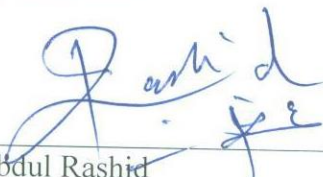
Supervisor:


Dr. Saud Ahmad Khan
Assistant Professor,
PIDE, Islamabad

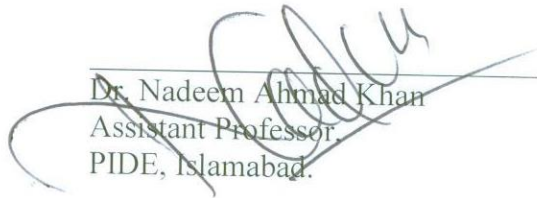
Internal Examiner:


Dr. Ahmad Fraz
Assistant Professor,
PIDE, Islamabad.

External Examiner:


Dr. Abdul Rashid
Associate Professor,
IIIE, IIU, Islamabad.

Head, Department of Business Studies:


Dr. Nadeem Ahmad Khan
Assistant Professor,
PIDE, Islamabad.

November 29, 2019

DECLARATION

I **Kaifayat Ullah**, student of M.Phil. Economics and Finance solemnly declare and affirm on oath that I myself have authored this M.Phil. thesis with my own work and means and I have not used any further means except those I have explicitly mentioned in this document. All items copied from internet or other written sources have been properly mentioned in the quotation marks and with a reference to the source citation.



Kaifayat Ullah

ACKNOWLEDGEMENT

Praise is due to Almighty ALLAH, the Compassionate and the Merciful, for His explicit and implicit favors. Peace and blessing are on the brilliant light, the giver of good tiding and warning, our master Muhammad and on his infalling Household and the righteous Companions.

I am highly obliged and grateful to my supervisor, monitor, and inspiring teacher professor Dr. Saud Ahmad Khan for his extended guidance, constant and unconditional encouragement that led my work to presented form. I am deeply indebted to my Ex-Head of Department Economics and Finance (PIDE) Dr Ahsan ul Haq Satti for his trust and support. I am also thankful to the staff at Department of Economics and Finance (PIDE).

I must thank my teacher and friends, for the contribution they made intentionally or unintentionally, scholar or just giving words of appreciation, moral support, courage and wisdom. I am grateful to Dr. Hafsa Hina and Dr Muhammad Javid for their support and valuable suggestion, and especially to my dear friends Mr. Amir Azam, Mr. Sikandar Hayat and Mr. Muhammad Saeed Meo for sharing their valuable knowledge of economics theory and econometrics and also provide me relevant material/information and assisting in typing, data collection, and estimation.



KAIFAYAT ULLAH

Table of Contents

DECLARATION	i
ACKNOWLEDGEMENT	ii
LIST OF TABLE	v
Abstract.....	vi
Chapter 1.....	1
INTRODUCTION	1
1.1 Background of the study	1
1.2 Problem statement.....	3
1.3 Research Gap:	4
1.4 Objectives of the study.....	4
1.5 Research Question	5
1.6 Significance of study.....	5
1.7 Organization of the Study	6
Chapter 2.....	7
REVIEW OF LITERATURE	7
Introduction.....	7
2.1 Theoretical Transmission Mechanism between Oil Price and Stock Market:	7
2.1.1 Stock Valuation Channel:	7
2.1.2 Monetary Channel:.....	8
2.1.3 Output Channel:	10
2.1.3 Fiscal Channel:.....	11
2.1.4 Uncertainty Channel:	11
2.2 Review of Empirical Literature.....	12
2.3 Literature Summary:	18
Chapter 3.....	19
METHODOLOGICAL FRAMEWORK	19
Introduction.....	19
3.1 Augmented Dickey Fuller Test.....	19
3.2 Econometric Specification of Non-Linear ARDL	20
3.3 Estimation Procedure of Non Linear ARDL.....	24

3.4 Diagnostic Test	25
3.5 Source of Dataset	26
3.6 Issues in Time Series Data	26
3.7 Description of the Variables	27
3.7.1 Real Economic Activity Index	27
3.7.2 World Crude Oil Production	28
3.7.3 Real Oil Prices	28
3.7.4 Volatility of Stock Price	28
CHAPTER 4	29
EMPIRICAL RESULTS AND DISCUSSIONS	29
4.1 Summary Statistics	29
4.2 Correlation Matrix	31
4.3 Augmented Dickey-Fuller Test Result (ADF-Unit Root Test)	32
4.4 Bound Test Result for Non-Linear Co-integration	33
4.5 Non-Linear ARDL Estimation Result	33
4.6 Wald Test result for Asymmetric Effect	37
4.7 Model Stability Test	37
CHAPTER: 5	39
CONCLUSION AND POLICY RECOMMENDATION	39
5.1 Policy Recommendation	40
REFERENCES	41

LIST OF TABLE

<i>Table No</i>	<i>Description</i>	<i>Page No</i>
<i>Table 1; Descriptive Statistics</i>		<i>29</i>
<i>Table 2: Correlation Matrix</i>		<i>31</i>
<i>Table 3: Unit Root Test Results</i>		<i>32</i>
<i>Table 4:Bound Test Result</i>		<i>33</i>
<i>Table 5: Nonlinear ARDL estimation results</i>		<i>36</i>
<i>Table 6: Wald Statistics Test Result</i>		<i>37</i>

Abstract

This thesis identifies different underlying oil price shocks to investigate the consequences of these shocks on the volatility of stock market prices in Pakistan which is highly import dependent countries. With some reasonable modification to the existing approach, this thesis uses Non-Linear Auto regressive distributed lag model to investigate the response of stock price volatility to oil price shock specifically oil demand shocks, oil supply shocks and oil specific demand shock by using monthly time series data from 2001-2018 for Pakistan. The analysis is carried out in two steps. In the first steps underlying source of oil price is identified. In the second step, the impact of these oil price shocks on the volatility of stock prices is analyzed for Pakistan. Bound test for nonlinear co-integration result indicate that there is long run association among the projected variables. The results of NARDL model is summarized as, except oil specific demand shock all variables have a long run and short impact on stock market volatility with different magnitude which may contribute in asymmetry. This asymmetry impact of projected variables is investigated with the help of WALD test. The result shows that in long run, oil supply shock, oil demand shock has an asymmetric effect on volatility of stock market whereas, oil specific demand shock test statistics indicate the absence of asymmetric effect. While in short run only oil demand show an asymmetric impacts and rest of the two variables namely oil supply shock and oil specific demand shocks results indicate that there is no asymmetric effect on volatility of stock prices.

Chapter 1

INTRODUCTION

1.1 Background of the study

Volatility is a measure of intensity of unpredictable changes in asset returns, It described by the continuous fluctuations in price and its volatility level would be described by its level of fluctuation (Adeniji 2014).Generally, it is linked with assets price which depict that to what and how regularly the price of assets will deviate significantly over the time. In opposite stock with fewer volatility have a price which stray comparatively little over the time. Volatility define the level of unpredictability or dispersion from a central point (Oluseyi 2015).In line with this (Abken and Nandi 1996) specify that movement in stock prices, future contract or any other market along with dispersion around mean and average return of assets is called a volatility. Volatility may affect the performance of economy due to its connectivity to the variance of assets price which may harm the financial system and smooth functioning (Mollah 2009). There is a belief behind the theory that oil prices are endogenous and innovations to both supply and demand motivated it (Bastian et al. 2016).However, oil market development specially rise in world oil prices will tend to reduce economic activities by production and consumer goods (petrol and heating oil) and producing volatility in financial market (Masih,et al 2011). Moreover, the fierce fluctuation in crude oil price will have significant effect on economy and its finance dividend discount model intended that for production crude oil is important factor in line with the real economic activities, which hits the production cost of enterprises, changing the profit margin with reflection in stock prices.

Being a developing economy Oil import is one of the major elements of Pakistan's import (Siddiqui, 2004). Therefore the economy is depending heavily on the oil as input in industrial, transport and power generation. Since many of the developing nations heavily oil important dependent and the change in prices of oil cause many of the structural and policy shocks in the economy. Specifically the return of stocks in developing economies has more fluctuations and the change in prices of oil makes them more fluctuating. There is fact that the changes in oil prices have potential of affecting the real economy showing that it may also have repercussions on the financial markets and there is an interaction exists between the performance of stock exchanges and the oil prices (Syzdykova, 2018). Usually the given interaction of oil prices and stock exchange returns depends on the dependency level of the affected economy to the oil and this show as the changes in the economical parameters.

Since our stock market is on the view of some internal and foreign economists and observer's view is also not stable i.e. Majidi, (2006) and Vera, (2010) propped that being a developing economy and highly depending on imports as compare to less diversified exports the stock market returns are so much volatile and fluctuating over the period and usually leave a debatable question mark on its success and failures. Moreover the industrial sector analysis have been confirmed a significant positive impact of oil prices on the firms' stocks returns while the oil price changes have been observed effecting the stock market returns negatively. Since industrial sector is consider as one of major instrument that let down the imports and promote export sector have been observed negatively affecting due to price changing of oil import.

To be specific the economy of Pakistan is facing significant impact of oil price change in both sectorial and structural system throughout its journey due to highly dependency of oil and different other energy sources. The small change in oil prices let the economy to suffer heat to

the cost of production that confine to earn less profit margins of corporation and business sector, reduce the consumer demand because of higher inflation in effect of increase in oil prices (Ansar and Asghar (2013) whereas the oil price shocks amend the uncertainty of investment create demand for higher rate of returns by investors and business sector that lead to constraining the level of investment at stock market and being already suffered stock market that inhibiting the level of investment in financial and real assets sectors leading to plummet the share prices in stock market due to nosedive in profits and higher discount rate Arshad and Bashir (2015). Despite the fact that evidence proposes that the origin of the oil price shocks triggers different responses from stock market and the majority of literature doesn't consider them when examines the effects. Hence through this study we will try to capture the impact of Oil Price shock on stock market volatility in Pakistan.

1.2 Problem statement

Developing countries heavily dependent on the import of oil which leaned them to face the socio economic and macro-economic problems. Pakistan is facing these problems through the decades due to his dependency on oil and other energy sources. This increase in oil price will give a heat to the cost of production ,which confine to decrease the profit margins of corporation, reduce the consumer demand because of increase in inflation in effect of increase in oil prices. Whereas, Oil price shocks amend the uncertainty for investment create a demand for higher rate of returns by investors which will lead to constraining the level of investment at stock market and inhibiting the level of investment in financial and real assets .This will lead to plummet the share prices in market due to nosedive in profits and higher discount rates.

Furthermore, financial data is usually high volatile for capturing the market volatility. Most of the researcher in this study shows that ARCH is founded in a high frequency data while in low

frequency data ARCH effect is fade out. In line with this several other studies have used GARCH family models to find out the impact of oil price shock on stock returns. In Pakistan the oil contracts are made monthly or fortnightly; means with high frequency. The contribution of this study is to estimate the impact of oil price shocks on volatility of stock in Pakistani stock market through Non-Linear ARDL model. So it is empirically found that ARDL modeling is easy to interpret and also address the ARCH effect in low frequency data. Therefore ARDL modeling is easy to estimate while GARCH models have convergence problems.

1.3 Research Gap:

Almost the whole world is depending on oil consumption and stock markets are backbone of countries investment scenario. There has been a lot of work done to check the impact of Oil Price Shocks both in Supply and demand sides using different econometrical and statistical approaches and have confronted different sorts of relationship with stock market returns. Pakistan being a developing economy mostly relies on Import of Oil and the stock market of the economy is also not well structured with several issues and challenges. The current study will highlight the impact of oil Price shocks on stock returns using Nonlinear Autoregressive Distributed Lag model that will show how the oil Price shocks affects the stock market returns in three different scenarios i.e. Oil Supply Shocks, Oil Demand Shocks and Oil Specific Demand Shocks.

1.4 Objectives of the study

Volatility affect assets return as its magnitude is judged by the level of its fluctuations. Countries dependent on oil imports will frequently observed the changes in assets return lead by the fluctuation at international market for oil, which contribute into deteriorate the economic growth. However, stock market is aligned to investors who make sell and purchase of different stocks on the basis of their future speculation. Therefore this study is being designed to meet the following

objective for assisting both policy makers and investors who can get plausible information regarding oil price shocks and stocks return.

1. To examine the impact of oil price shocks (supply-side shocks, demand-side shocks and oil specific demand shocks) on stock market prices volatility in Pakistan.
2. To examine the asymmetric cumulative multipliers of oil prices shock and stock market prices.

1.5 Research Question

On the base of above research gap, this study is trying to answer these two question. The questions are following:

1. What is the impact of oil price shock specifically oil demand shocks, oil supply shocks and oil specific demand shock on stock price volatility?
2. Weather stock price volatility show asymmetric response to oil price shocks?

1.6 Significance of study

This study utilizes non-linear autoregressive distributed lag approach to investigate long run and short run relationship between oil price shocks specifically oil demand shocks, oil supply shocks, oil specific demand shock and volatility of stock price. This study will help investor in their decision making regarding the formulation of investment portfolio. Both type of investor including day trader (overnight trader) and short term investor can formulate their portfolio in the light of short run result of this study. Furthermore, this study will also help those investors who are investing for long time period. Normally, long term investors invest in those stocks whose return hold law of average.

1.7 Organization of the Study

First chapter of this study is introductory part. Chapter 2 of this study is related to prior theoretical and empirical works. Third chapter of this study is discussing methodological framework, explanation of the variable, econometric model and estimation technique of the model. Chapter 4 of this study is about result and discussion. Finally, last chapter of this study consist of conclusion and policy recommendation.

Chapter 2

REVIEW OF LITERATURE

Introduction

This chapter focuses on the theoretical background and previous literature associated with the primary objective of our study, namely impact of oil price shocks on stock market volatility in Pakistan. Important studies are discussed in this chapter in order to establish the link between oil prices shocks and stock market volatility. First section of this chapter deals with oil price Transmission Mechanisms while second section of the study discuss previous research done and the later part of this chapter discussed the summary of the chapter.

2.1 Theoretical Transmission Mechanism between Oil Price and Stock Market:

In this section we set the scene and explain some theoretical transmission mechanisms by which oil price changes can alter the behavior of stock markets. We categorize the channels in five different ways.

2.1.1 Stock Valuation Channel:

The stock valuation channel is the direct channel by which oil prices influence stock markets. Making this clear requires two equations: first, we define stock returns ($R_{i,t}$) as the first log-difference as:

$$R_{i,t} = \log(P_{i,t}/P_{i,t-1}) \dots \dots \dots (1)$$

Where ($P_{i,t}$) denotes the stock price of firm at time. Second, economic theory suggests that current stock prices reflect the discounted future cash flows of a particular stock (Huang et al., 1996). This can be shown as:

$$P_{i,t} = \sum_{n=t+1}^N \left(\frac{E(CF_n)}{(1 + E(r))^n} \right) \dots \dots \dots (2)$$

Where CF_n is the cash flow at time n and r is the discount rate. $E(.)$ denotes the expectation operator. Eqs. 1 and 2 show that stock returns are impacted by factors that can alter the expected cash flows and/or the discount rate, including oil prices. Oil price changes can alter a firm's future cash flows either positively or negatively, depending on whether the firm is an oil-user (oil-consumer) or oil-producer (see Oberndorfer, 2009; Mohanty and Nandha, 2011). For an oil-consuming firm, oil is one of the major production factors and consequently an increase in oil prices will result in an increase of production costs (assuming that there are no perfect substitution effects between production factors, see Basher and Sadorsky, 2006), which, in turn, will reduce profit levels and thus future cash flows (Bohi 1991; Mork, Olsen, and Mysen 1994; Hampton, 1995; Brown and Yucel 1999; Filis et al., 2011). On the other hand, for an oil-producer the oil price increase will result in increased profit margins and thus increased expected cash flows. Intuitively, we expect oil-users to exhibit bearish behavior during periods of oil price increase, whereas the reverse holds true for oil producing firms.

2.1.2 Monetary Channel:

Oil price changes also affect the expected discount rates of future cash flows (see Eq. 2). According to Mohanty and Nandha (2011), the discount rate is at least partially composed of expected inflation and expected real interest rates. Thus, the second transmission mechanism by

which oil price changes impact stock returns is through inflation and interest rates. As mentioned in section 2.1, rising oil prices result in increased production costs. However, these costs will be transferred to consumers, leading to higher retail prices and thus higher expected inflation (see Abel and Bernanke 2001; Hamilton 1996, 1988; Barro 1984, among others). Assuming that a central bank follows some type of rule 2.1, we expect monetary policy makers to increase short-term interest rates in response to higher inflationary pressures (Basher and Sadorsky, 2006). There are two main effects of the increased short-term interest rates on stock markets. First, increases in short-term interest rates lead to an increase in commercial borrowing rates (i.e., discount rates) for any future firm investments, raising the borrowing costs of firms. Furthermore, the increased borrowing costs lead to fewer positive NPV projects (lower cash flows). Thus, either due to increased discount rates and/or lower cash flows, stock prices decrease in value.

We should highlight here that the magnitude of the aforementioned effects depends on the central bank's credibility to stabilize inflation. Assuming a highly credible central bank, we maintain that inflation expectations will remain stable, despite an oil price increase, and thus close to the inflation target. Through this expectations channel, we do not expect a significant increase in inflation following an oil price increase. By contrast, in the case of a low credibility central bank, inflation expectations will be volatile and this results in a larger change of inflation expectation, following an oil price increase, leading to an even worse impact on stock price levels.

2.1.3 Output Channel:

The third channel is the output channel. The literature maintains that oil price fluctuations affect aggregate output (see, *inter alia*, Hamilton, 1983; Hamilton, 2003; Kilian, 2008a, 2008b; Hamilton, 2009a). According to this channel, positive oil price changes are expected to have both an income and a production cost effect, which will lead to changes in aggregate output. The production cost effect was explained in section 2.1, so we will concentrate on the income effect in this section. More specifically, increased oil prices tend to lead to lower the discretionary income of households, due to the changes in retail prices (as a result of increased production costs), but also due to the increased prices of gasoline and heating oil (Bernanke, 2006; Edelstein and Kilian, 2009). Lower income leads to lower consumption and thus aggregate output, which further leads to lower labour demand. Put differently, an increase in oil prices will worsen the terms-of-trade for an oil importing economy, which will result in lower income and a negative wealth effect on consumption, and in turn to lower aggregate demand (Svensson, 2005 and 2006). Stock markets tend to respond negatively to such developments. We maintain that this will be the response of stock markets, based on Equ 1 and 2. In particular, lower aggregate demand leads to lower expected cash flows for firms, which further leads to lower stock prices. These effects are not expected to hold for all economies. On the contrary, they depend on whether an economy is oil-importing or oil-exporting. The aforementioned sequence of events holds for an oil-importing economy. On the other hand, even though an oil-exporting economy will also experience negative production cost effects, it will benefit from a positive income effect, due to increased oil revenues (the value of export demand for oil rises), leading to higher aggregate demand and thus higher output. The positive change in the aggregate demand will occur only if the income effect is such that it can counterbalance the negative production cost

effect. In such a case, stock markets will respond favorably to the increased output, as it will boost the expected cash flows of the firms that operate in the country.

2.1.3 Fiscal Channel:

The fiscal channel is primarily concerned with oil-exporting economies, which are financing physical and social infrastructure using their oil revenues (see, Ayadi 2005; Farzanegan 2011; Emami and Adibpour 2012). Increased oil prices tend to lead to a transfer of wealth from oil-importing economies to oil-exporting ones (Dohner,1981), which allow for increased government purchases. Assuming that consumption and government purchases are considered complements, then the latter will lead to higher household consumption. In such a case, private firms are expected to increase their cash flows and thus their profitability. Such developments will push stock prices to higher levels and the stock market will exhibit a bullish period. By contrast, if consumption and government purchases are regarded as substitutes then the opposite impact will be evident, due to the crowding out effects. Stock markets will respond negatively to such developments, as the substitution effect will drive out the most productive private capital of the economy.

2.1.4 Uncertainty Channel:

The final transmission channel is the uncertainty channel, suggested by Brown and Yücel (2002). In particular, rising oil prices cause higher uncertainty in the real economy, due to the effects of the former on inflation, output, consumption, etc. Thus, increased oil prices will reduce firms' demand for irreversible investments, which in turn, reduce expected cash flows. Furthermore, uncertainty is also propagated to households which reduce their consumption of durable goods (Bernanke 1983; Pindyck 2003). Rising uncertainty about future oil costs increases the incentives of households to save rather than consume (Edelstein and Kilian, 2009). It is worth

noting here that as uncertainty rises due to increased oil prices, the value of postponing both investment and consumption decisions increases and thus, a decrease in the incentive to invest or consume is observed, which thereby dampens economic growth prospects (Chuku et al., 2010) and thus stock market returns.

2.2 Review of Empirical Literature

(Sadorsky 1999) have gone through an analysis over the impact of oil price shocks on stock market activities with the help of GARCH models to model the oil price volatility come to end with findings that both oil prices and the volatility of oil prices put a significant impact over stock market activities and their returns.(Ratti and Hasan 2013) using the GARCH and GARCH in mean models have tried to examine the effect of oil price return and oil return volatility at a sector wise for Australian stock market to comprehend the risk and return tradeoff of these sectors concluded that a rise in oil price returns will have a significant impact on returns and the volatility of this oil price return will significantly reduce the volatility.

(Park and Ratti 2008) produced the study to seen out the relationship between stock market and oil price shock for US aligned with 13 other European countries by approaching the widely used technique variance decomposition analysis in respect to that he finalized that Norway stock return retort positively to oil price increases. Adding further he revealed that oil price shocks account 6% of the volatility in stock return which may strengthen the claim that volatility will significantly weaken the stock return for the above mentioned countries expect US due to greater changes in return as compared to interest rate.

The formative paper produced by (Jones and Kaul 1996)was first to delineate the negative relationship between oil prices and stock returns. Following this, Sadorsky (1999) concludes that

oil prices are strong determinants of stock market returns specifically; stock market returns retort negatively to the positive amendments in oil prices, beside these there are too many papers including (Filis 2010).(Miller and Ratti 2009) who supports the negative relation between the two.

However, above negative relation does not hold the water for stocks functioning in export oriented countries, (Arouri and Rault 2012) confirm the positive relation between oil price shock and stock return. A strand of studies have differentiated the effects in line with their origin (Hamilton 2009) and (Kilian,(2008) improvised that different shocks have a different impact on stock markets. (Kilian (2009) present the evidence that in respect to the cause oil prices shock respond different to stock returns. Hamilton (2009 a,b) have disaggregated the oil price shocks into two components demand side shocks and supply side shocks . Furthermore, Kilian (2009) have identified the third origin oil specific demand shocks which may arise due to the uncertainty in future accessibility of oil.

Hu et al (2017) studied the long run and short relationship asymmetric effects to check the impact of structural oil price shocks on the Chinese stock market and the findings of the study suggests that there is no significant impact of supply and demand shocks but in oil specific demand there has been observed asymmetric effects of oil prices on stock market where the supply, demand and demand specific shock been derived using SVAR model and the impact of these shocks has been analyzed through NARDL Approach.

Al-hajj et al (2018) analyzed the sectorial impact of Oil Price shocks in Malaysia by focusing the valuation in oil prices with different control variables using NARDL approach on a time panel data of 1990-2016 concluded that there is inverse relationship between Oil Price shocks and stock market returns by keeping the appreciation and depreciation of Oil Prices and there

exists strong long run asymmetric between Oil Price shocks and Stock market returns with certain other variables in the study.

Kang et al, (2015) examined the impact of structural oil price shock on the co-variance of United State's stock market return and volatility using SVAR concluded that the positive shocks in aggregate demand and oil market specific demands are closely related with negative impact on the covariance return and volatility while supply shocks are associated with positive effects on covariance of return and volatility with a very high significant level.

Syzdykova (2018) analyzed the relationship between crude oil price shocks and stock prices in Common Wealth economies by using ARDL co-integration approach from a time period 2010 - 2017 and concluded that crude price shocks and the fluctuations in stock prices are co-integrated and there exists long run relationship between stock prices and changes in oil prices with Industrial Production Index and interest rate as control variable while the speed of adjustment in long run is asymmetrical in behavior.

Degiannakis et al (2014) analyzed panel data of European region to check the relationship between oil price shocks and stock market volatility, based on conditional, implied and realised volatility confronted that supply side shocks demand specific shocks have no significant impact on stock market volatility while the aggregate demand shocks lead to reduction in stock market volatility that are caused due to oil price changes.

Marashdeh & Afandi (2017) concluded in their study on the Oil Price Shocks in Oil Producing countries that the small economies according to GDP get negative significant impact of oil price shocks as compare to those whose GDP is higher than that i.e. The oil price shocks rising in USA will have significant impact on the stock market of Saudi Arabia while the supply or

demand shocks of Saudi Arabia have no significant impact on the USA stock market returns in short run while in long run it has significant negative impact on the stock return.

Broadstock & Filis (2014) made comparative analysis of Oil Price shocks and stock market return between USA and China and used several econometrical and statistical techniques and confronted that there is exists time varying systematic correlation between oil price shock and stock market of both economies while the strength of impact differs between the economy where the Chinese economy is more vulnerable as compare to USA because the resilience of Chinese economy is more than that of USA.

Naurin and Qayyum (2016) target to explain the impact of oil price and its volatility on stock market index in Pakistan. Their finding of bi-variate EGARCH model suggested that there is a direct association between oil prices and stock market index.

O'Neil et al (2008) argued the negative impact of oil price shocks especially on output and profitability of oil intensive firms. Andreas and Constantinos (2009) explored the effects of oil price returns on oil price volatility of countries like Greece, the US, the UK and the German stock markets using EGARCH models and structural equation models and concluded that Greek and US stock market returns are both sensitive to the oil price returns movements while the German and the UK stock market returns are not affected at all. Kilian (2009) criticized the past studies, because researchers treated oil-price shocks as exogenous. There are studies on the other hand, which argued that oil price changes also react to the same causes affecting the stock market prices also (Hamilton, 2005; Kilian, 2009). Therefore, aggregate oil price shocks must be decomposed into the underlying structural shocks reflecting the endogenous features of such oil price shocks. Masih et al (2010) accessing emerging market like South Korea economy explored the impact of oil price volatility on stock price fluctuations though model was formed which

includes variables such as interest rates, economic activity, real stock returns, real oil prices and oil price volatility by using co-integration test, VEC model, variance decomposition and impulse responses and the findings of co-integration test showed a long run relationship among the variables and VEC model results revealed the supremacy of oil price volatility on real stock returns and accentuate how this has increased over time with conclusion that oil price volatility can have profound effect on the time horizon of investment and firms need adjust their risk management procedures accordingly.

Ono (2011) examined the impact of oil prices on real stock returns for different countries using VAR mode and his result revealed that oil price volatility has positive and significant impact on real stock returns of countries like China, India and Russia, while the reverse is the case for that of Brazil. It was also profound that for a country like India, statistically significant asymmetric effects of oil price increases and decreases are observed, and the analysis of variance decomposition result showed that the contribution of oil price shocks to volatility in real stock returns is relatively large and statistically significant for China and Russia. Oskooe (2011) investigated the dynamic relationship between stock market and oil price in an oil-exporting country (Iran) and international oil market using mean and variance techniques of analysis. The findings from the analysis revealed that, the variance of oil price fluctuations does not cause the variance of Iran stock returns and it was concluded that there is no volatility spillover effect between Iran stock market and international oil market.

Hasan and Ratti (2012) examined the effect of oil price return and oil return volatility on the return and volatility of the sectors of Australian stock market respectively and study the risk-return trade off of these sectors. They employed the GARCH-in-mean (GARCH-M) methodology and found that for the overall market index, an increase in oil price return

significantly reduces return, and an increase in oil price return volatility significantly reduces volatility. Specifically, for eight out of ten sectors (significantly so for six sectors) oil price return and stock price return move in opposite directions, but for the energy and materials sectors increased oil price return increases sector returns. Chen, Hamori and Kinkyo (2014) made an extension to the Kilian's (2009) model by identifying an exogenous underlying shock to the oil prices, which arises from fluctuations in the financial market conditions and examined the resultant impacts of oil price changes on the macroeconomic variables. Their results confirmed the important role of financial shock in determining the oil prices and consequently on macroeconomic changes. Fatima and Bashir (2014) investigated the relationship between oil prices volatility and stock market changes in Pakistan and China. They found evidence in favors of negative and asymmetric effect of oil price on these emerging stock markets. Effiong (2014) provided the study which examined the impact of the origin of oil price shocks on Nigeria's stock market where he decomposed oil price shocks into oil supply shocks, aggregate demand shocks and oil-specific demand shocks and employed a structural vector auto-regression model (SVAR), impulse response and variance decomposition and the results from the impulse response revealed that, stock market's response to oil supply shocks is insignificantly negative but significantly positive to aggregate demand and oil-specific demand shocks and the collective effects of the oil price shocks account for about 47 per cent of the variation in stock prices in the long term. His results advised that the source of oil price shocks is essential for understanding the volatility in Nigeria's stock market Ekong and Effiong (2015) investigated the effects of demand and supply shocks in the crude oil-market on Nigeria's macro economy. They employed a structural vector autoregressive (SVAR) model, while oil price shocks were disentangled into three components: oil supply shocks, aggregate demand shocks and oil-specific demand shocks. The result from the

impulse response revealed that, the macroeconomic aggregates are associated with different response pattern to each type of oil price shocks. However, their relative importance shows marginal effects. Lawal, et al (2016) examined the impact of both the exchange rate volatility and oil price volatility on stock market volatility in Nigeria. They employed EGARCH estimation techniques to ascertain if either the volatility in exchange rate, oil price volatility or both exerts on stock market volatility in Nigeria. Their result exposed that share price volatility is induced by both the exchange rate volatility and oil price volatility and it was recommended that policymakers should pursue policies that tend to stabilize the exchange rate regime on the one hand, and guarantee the net oil exporting position for the economy, that market practitioners should formulate portfolio strategies in such a way that volatility in both exchange rates and oil price will be factored in time when investment decisions are being made. In a most recent study Jebran et. al. (2017) reported adverse effects of oil price shocks in periods before the 2007 financial crisis where as positive effects in periods after such crisis. Bastianin and Manera (2017) while examining the reaction of U.S stock market volatility to the oil price shocks also used the SVAR method. They found results in favor of important role of aggregate demand shocks and oil specific demand shocks whereas less important role of oil supply shocks.

2.3 Literature Summary:

From the above theoretical and empirical studies around the world and Pakistan we can see that most of the works on the relationship between Oil Price shock and stock market volatility of Pakistan have been conducted through using Linear Analysis and Structural Vector Autoregressive method. In the current study we will focus on Non Linear Analysis technique for decomposition of oil Price shock on Stock Market Volatility.

Chapter 3

METHODOLOGICAL FRAMEWORK

Introduction

It is of immense importance to empirically test the theoretical underpinnings. Without empirical testing, one can't be sure that whether theory is correct or not. In this thesis, the author plan to investigate the impact of oil demand shocks, oil supply shock and oil specific demand shock on stock price volatility in the long run as well as in short run. To estimate the model, one need data and has to specify estimation technique. In this chapter such issues are discussed in detail.

This chapter presents an extensive overview of the order of integration of the variables, econometric specification non-linear autoregressive distributed lag (NARDL) model followed by diagnostics tests, explanation of variable, stability of the estimated parameter, cumulative asymmetric effect and the source of the dataset. The later part of this chapter discusses issue in time series data.

3.1 Augmented Dickey Fuller Test

$$\Delta y_t = \alpha + \beta y_{t-1} + \mu_t \dots (3.1)$$

Where $\Delta y_t = \text{first difference of timeseries}$ and the unit root coefficient $\beta = \delta - 1$. so the null hypothesis is

$$H_0 : \beta = 0 \text{ or } \delta = 1$$

And the alternative hypothesis would be

$$H_a : \beta < 0.$$

The error term μ_t is assumed to be purely white noise. Generally the error term is not white noise so the dickey and fuller suggested another form of above equation which controls for the problem of autocorrelation. Augmented term is simply the lag of dependent variable. The lag length is decided by AIC and BIC criterion. There are three specification of this test.

$$\Delta y_t = \gamma y_{t-1} + \sum_{i=1}^p \beta_i \Delta y_{t-i} + \mu_t \dots \quad (3.2)$$

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^p \beta_i \Delta y_{t-i} + \mu_t \dots \quad (3.3)$$

$$\Delta y_t = \alpha + \gamma y_{t-1} + \alpha_2 t + \sum_{i=1}^p \beta_i \Delta y_{t-i} + \mu_t \dots \quad (3.4)$$

These three specifications are different from each other with respect to deterministic elements α and α_2 . The selection of suitable model would be on the basis of plotting the graph of series and observing whether there is deterministic component or not and then choosing model accordingly.

3.2 Econometric Specification of Non-Linear ARDL

Non-linear ARDL model in recent times was advanced by Shin, Yu, and Greenwood-Nimmo (2014). The method employs positive and negative partial sum decompositions for discovering the asymmetric effects both in the long and the short-term. Unlike the classical co-integration models, NARDL models produce better result when the long run relationship is sort for in variables with small sample size (Romilly, Song & Liu 2001), it can be applied to variables that are stationary at level $I(0)$ or at first difference $I(1)$ except for variables that is stationary at second difference $I(2)$. Specifically, asymmetric NARDL framework of Shin et al (2013) is

appropriate for this study because of its ability to reveal the short- and long-run asymmetries and still spot concealed co-integration. This will allow variation in the effect of a positive and negative oil price shocks in the short-run and long-run.

Now, to investigate the impact of oil price shock on the volatility of stock prices in case of Pakistan. We are following the procedure of Effiong (2014) to meet the primary objective of this study. We developed the following linear model,

$$PSX_t = f(OSS_t, ODS_t, OSDS_t) \dots \dots \dots (3.5)$$

Where, PSX shows Pakistan stock market prices. While, OSS, ODS, OSDS represent oil supply shock, oil demand shock and oil specific demand shocks respectively. Based on the statement of Engle (2002), who suggested that the residual of autoregressive process is accountable to reveal volatility more than any other method. GARCH variance series (VPSX) will be generated and will serve as the pure volatility (with neither exogenous nor endogenous) of PSX. Hence, equation 3.5 can be written as

$$VPSX_t = f(OSS_t, ODS_t, OSDS_t) \dots \dots \dots (3.6)$$

The equation 3.6 shows the linear impact of independent variable on stock price volatility. While the objective of this study is to investigate the nonlinear impact of oil demand shocks, oil supply shocks and oil specific demand shock on the stock price volatility. To achieve the objective of this study, we decompose the oil demand, oil supply and oil specific demand into positive and negative shocks. Hence equation 3.6 can be written as

$$VPSX_t = \delta_o + \delta_1 OSS_t^+ + \delta_2 OSS_t^- + \delta_3 ODS_t^+ + \delta_4 ODS_t^- + \delta_1 OSDS_t^+ + \delta_2 OSDS_t^- + \mu_t \dots \dots \dots (3.7)$$

Equation 3.7 depicts the nonlinear relationship between volatility of stock prices and the oil supply shocks, oil demand shocks as well as oil specific demand shocks in long run. Where δ is the vector of long term parameters,

$$\delta = (\delta_0, \delta_1, \delta_3, \delta_4, \delta_5, \text{ and } \delta_6)$$

From equation 3,7 the partial sum of positive and negative changes in oil supply shocks can be calculated as

$$OSSt^+ = \sum_{i=1}^t \Delta OSSt^+ = \sum_{i=1}^t \max(\Delta OSS_i, 0) \quad (3.8a)$$

$$OSSt^- = \sum_{i=1}^t \Delta OSSt^- = \sum_{i=1}^t \min(\Delta OSS_i, 0) \quad (3.8b)$$

However, for oil demand shocks the partial sum of positive and negative changes could be estimated by

$$ODSt^+ = \sum_{i=1}^t \Delta ODSt^+ = \sum_{i=1}^t \max(\Delta ODS_i, 0) \quad (3.9a)$$

$$ODSt^- = \sum_{i=1}^t \Delta ODSt^- = \sum_{i=1}^t \min(\Delta ODS_i, 0) \quad (3.9b)$$

However, for oil specific demand shocks the partial sum of positive and negative changes can be computed as

$$OSDSt^+ = \sum_{i=1}^t \Delta OSDSt^+ = \sum_{i=1}^t \max(\Delta OSDS_i, 0) \quad (3.10a)$$

$$OSDSt^- = \sum_{i=1}^t \Delta OSDSt^- = \sum_{i=1}^t \min(\Delta OSDS_i, 0) \quad (3.10a)$$

We can rewrite equation 3.7 into ARDL form as thus

$$\begin{aligned}
\Delta VPSX_t = & \delta + \delta_0 VPSX_t + \delta_1 OSS_{t-1}^+ + \delta_2 OSS_{t-1}^- + \delta_3 ODS_{t-1}^+ + \delta_4 ODS_{t-1}^- + \delta_5 OSDS_{t-1}^+ \\
& + \delta_6 OSDS_{t-1}^- + \sum_{l=1}^{\alpha} \alpha VPSX_{t-l} + \sum_{i=0}^b (\beta i^+ \Delta OSS_{t-1}^+ + \beta i^- \Delta OSS_{t-1}^-) \\
& + \sum_{i=0}^c (\gamma i^+ \Delta ODS_{t-1}^+ + \gamma i^- \Delta ODS_{t-1}^-) + \sum_{i=0}^d (\vartheta i^+ \Delta OSDS_{t-1}^+ + \vartheta i^- \Delta OSDS_{t-1}^-) \\
& + \varepsilon_t
\end{aligned} \tag{3.11}$$

However, a, b, c and d are the lag orders whereas the long run impacts of supply shocks can be computed as

$$\delta_1 = \frac{-\delta_1}{\delta_0}, \delta_2 = \frac{-\delta_2}{\delta_0}$$

State that an increase in oil supply shock will reduces the stock prices. Moreover, the long run impacts of the demand shocks can be calculated as

$$\delta_3 = \frac{-\delta_3}{\delta_0}, \delta_4 = \frac{-\delta_4}{\delta_0}$$

In the same way, it can be interpreted as increase in oil demand shocks will decrease the stock prices. Finally, the long run impacts of oil specific demand shocks can be computed as

$$\delta_5 = \frac{-\delta_5}{\delta_0}, \delta_6 = \frac{-\delta_6}{\delta_0}$$

Similarly, the short run impacts are calculated by following method.

$\sum_{i=1}^b \beta i^+$, measure the short run impact of increase in oil supply shocks and $\sum_{i=1}^b \beta i^-$ measure the short run impact of decrease in oil supply shocks. Next, $\sum_{i=1}^c \beta i^+$ measure the short run impact of increase in oil demand shocks while $\sum_{i=1}^c \beta i^-$ measure the short run impact of decrease

in oil supply shocks. In the same way, $\sum_{i=1}^d \vartheta i^+$ measure the short run impact of increase in oil specific demand shocks and $\sum_{i=1}^d \vartheta i^-$ measure the short run impact of decrease in oil specific demand shocks.

For the calculation of asymmetric cumulative multiplier, we use the following formula

$$Ml^+(OSS) = \sum_{i=0}^l \frac{\partial VPSX_{t+1}}{\partial OSS_{t-1}}, l = 0.1.2.3.4 \dots \dots \dots (3.12a)$$

$$Ml^-(OSS) = \sum_{i=0}^l \frac{\partial VPSX_{t+1}}{\partial OSS_{t-1}}, l = 0.1.2.3.4 \dots \dots \dots (3.12b)$$

$$Ml^+(ODS) = \sum_{i=0}^l \frac{\partial VPSX_{t+1}}{\partial ODS_{t-1}}, l = 0.1.2.3.4 \dots \dots \dots (3.12c)$$

$$Ml^-(ODS) = \sum_{i=0}^l \frac{\partial VPSX_{t+1}}{\partial ODS_{t-1}}, l = 0.1.2.3.4 \dots \dots \dots (3.12d)$$

$$Ml^+(OSDS) = \sum_{i=0}^l \frac{\partial VPSX_{t+1}}{\partial OSDS_{t-1}}, l = 0.1.2.3.4 \dots \dots \dots (3.12e)$$

$$Ml^-(OSDS) = \sum_{i=0}^l \frac{\partial VPSX_{t+1}}{\partial OSDS_{t-1}}, l = 0.1.2.3.4 \dots \dots \dots (3.12f)$$

There for equation 3.11 to 3.16 can be summarized as

$$l \rightarrow \infty, Ml^+(OSS) \rightarrow \alpha_1, Ml^-(OSS) \rightarrow \alpha_2, Ml^+(ODS) \rightarrow \alpha_3, Ml^-(ODS) \rightarrow \alpha_4, Ml^+(OSDS) \rightarrow \alpha_5, Ml^-(OSDS) \rightarrow \alpha_6$$

3.3 Estimation Procedure of Non Linear ARDL

Following steps are involved in the estimation of nonlinear Auto regressive distributed lag model. The order of integration of the variables is analyzed in the first step in order to assure the absence of 2nd order of the integration of the variable because in the presence of 2nd order of

integration of the variable, the value of F-statistics of the bound test become invalid and the estimation result become insignificant. Furthermore, it is the advantage of ARDL framework that it can be applied even if all the variables are integrated at level or 1st difference or the mixture of level and 1st difference. For this purpose, this study utilize ADF unit root test and the results of ADF test will be confirmed by Phillip-Peron unit root test.

In the 2rd step, we estimate equation 3.10, with the help of standard ordinary least square method by using general to specific approach previously adopted by Katrakilidis and Trachanas (2012) and Ibrahim (2015) after the asymmetric variables have been disentangled into their respective form. This process involves continuous extraction and deleting of insignificant lag until a parsimonious result is reached. Furthermore, in the next step, bound test of co-integration developed by Pesaran, Shin & Smith (2001) and extended by Shin et al (2014) is performed to investigate the long run association among the projected variables. Following are the hypothesis of bound test of co-integration,

$$H_0; \delta_1 = \delta_2 = \delta_3 = \delta_4 = \delta_5 = \delta_6 = 0$$

$$H_a; \delta_1 = \delta_2 = \delta_3 = \delta_4 = \delta_5 = \delta_6 \neq 0$$

If the value of bound test F-statistics exceeds the tabulated value then we accept alternative hypothesis and reject the null hypothesis by concluding co-integration among the projected variables. After concluding co-integration among the projected variables we test for asymmetric effect of oil demand shocks, oil supply shocks and oil specific demand shocks on volatility of stock prices in both short run and long run.

3.4 Diagnostic Test

Furthermore, this model will be tested for autocorrelation, heteroskedasticity and normality issue of the residual, functional form of the model with the help of Breusch-Godfrey Serial Correlation

LM Test, Breusch-Pagan Godfrey Heteroskedasticity Test Jarque-Bera Normality Test and Ramsey Rest test respectively. The parameter stability will be diagnosed with the help of Cusum & Cusum of square test.

3.5 Source of Dataset

This study uses monthly dataset of stock prices, real oil prices, real economic activities index and crude oil production over the period of 2001 to 2018. The historical series of stock prices is based on the Pakistan stock exchange 100 index (PSX-100), while crude oil production and real oil prices historical series is obtained from US energy information administration (EIA). Furthermore, this study utilize real economic activities index obtained from homepage of Kilian's (2009).

3.6 Issues in Time Series Data

This study will use time series dataset to investigate the relationship between Oil Price Shocks Specifically oil specific demand shocks, oil demand shocks, oil supply shocks and volatility of stock market prices in context of Pakistan over the period of January 2001 to December 2018. Generally, time series data are secondary in nature and easy to collect. However, time series analysis has many problems such non stationarity of series, collinearity among the variable etc. Non stationarity of the series indicates that mean and variance of the not constant over time. Non stationarity of the series effect the analysis by giving spurious regression, which leads to biased estimate and result are not reliable. To deal with such problem we carried out the following remedial measure,

1. Descriptive statistics summary is calculated to know about the structure of dataset. Descriptive statistics will fulfill our objective such as distribution of dataset, helps to detect outlier in the dataset etc.
2. Stationarity of the series is investigated with the help of unit root test. For this purpose, this study utilizes ADF test. The result of Augmented Dickey Fuller (ADF) is confirmed by Phillip-Perron (PP) test.
3. On the base of unit root test results, proper econometric technique is selected for estimation of the model.

3.7 Description of the Variables

This section discusses all the variables used in this study in details. Explanation of the variables is necessary because it provide important information such as reason of selecting specific variables for this study, expected sign of the coefficient and its unit etc.

3.7.1 Real Economic Activity Index

World real economic activities index is used as a proxy for global oil demand for industrial commodities, because world real economic activities is based on dry cargo single voyage ocean freight rates of different commodities including grain, oil seeds, coal, iron, ore, fertilizer and scrap metal etc. world real economic activities index capture the shift in global oil demand in the world market. The index is deflated by US consumer price index and linearly de-trended to remove the effect of technological advances in ship building and other long term trend in the demand for sea transport. Therefore, the de-trended index captures the effect of cyclical variation in ocean freight rates (Kiliian, 2009).

3.7.2 World Crude Oil Production

Global oil production is used as a proxy for world oil supply. World oil production reflects both OPEC and non OPEC countries political uncertainties and cartel activities. Oil supply shocks capture unpredictable innovations in world crude oil production. We estimated oil supply shock from crude oil production series.

3.7.3 Real Oil Prices

Real oil prices are measured by using the free on board (FOB) importing price in the US by following Kolodzeji and Kaufmann, (2014). Free on board importing price of oil contains the effect of freight charges. In order to remove the freight rate effect, we deflated the free on board importing price of oil by US consumer price index which is available on IFS (International Financial Statistics).

3.7.4 Volatility of Stock Price

Volatility is pace at which stock price move up and downward and how wildly swing. It is statistical approach to measure dispersion in stock market index. Volatility has been most exhaustive studied measured and described in stock market. This study use volatility of stock price of Pakistan stock exchange (psx-100 index) as a proxy for market response. Furthermore there are different method to calculated volatility of stock prices including beta coefficient, option pricing and standard deviation approach. Each method has their own advantages and disadvantage but this study will follow standard deviation approach.

CHAPTER 4

EMPIRICAL RESULTS AND DISCUSSIONS

After the chapter of methodology, the result on the said methodology will be presented in this chapter. This chapter comprises of the major findings and the discussion is provided in reference to concerning the previous studies and theories. Results includes unit root test, lag length criterion to choose the optimal lag selection. Furthermore, this chapter gives the results of both ARDL and NARDL technique. Based on these findings we conclude the results.

4.1 Summary Statistics

The summary statistics mean, standard deviation, minimum, maximum, skewness and kurtosis of all the variables used in study for data from 2001-2018 are calculated, and the results obtained are mentioned in table 4.1.

Table 1: Descriptive Statistics

<i>Variable</i>	<i>Observation</i>	<i>Mean</i>	<i>Std.Deviation</i>	<i>Skewness</i>	<i>Kurtosis</i>
Volatility of Stock Price (VSP)	216	5158.204	7840.170	-0.398211	1.847264
Crude Oil Production (COP)	216	75122.99	4375.399	0.022197	2.397407
Real Economic Activity (REA)	216	10.24966	75.43235	0.4182252	2.397322
Oil Prices (OP)	216	66.5662	30.34501	0.632265	4.99850

Source; Author Own Computation

The descriptive statistics are used to describe the basic feature of the data used in this study. The descriptive statistics tell us “what the data shows”. Mean is one of the important estimates to

describe central tendency of the series. Mean give us the single value that describes the center of the series. Crude oil production has the largest mean value of (751122.99) followed by the average value of stock price volatility (5158.204). This indicates that the average production of world Crude oil was 75122.99 thousand barrels per month and on average volatility of stock price was (5158.204) during the study period ranging from 2001M1 to 2018M12. Furthermore, the average value of real economic activity index was 10.24966 and average price of crude oil was 66.5662 during the study period.

Standard deviation is most common technique to measure dispersion in the data. Standard deviation shows spread in the data from the mean, greater value of standard deviation indicate greater spread in the data and vice versa. Volatility of stock price got the highest value of (7840.170) and followed by the crude oil production having value of (4375.399). The lowest value of standard deviation is associated with oil prices which is (30.34501), followed by the real economic activity (75.83235). Another important measure of descriptive statistics is skewness which tells us about the symmetrical distribution. The value of skewness approach to zero means that data are more symmetrical. From the skewness column it can be easily concluded that all the projected variables are positively skewed except volatility of stock price. Next come to the kurtosis column which tells us about “how the peak and tails differ from normal distribution”. Generally, the kurtosis gives us understanding about the distribution of the data. The value of the kurtosis approach to zero (0) shows that data are normally distributed while positive value indicate that distribution has heavier tails and sharper peaks from normal distribution. Negative value of kurtosis indicates that distribution has flatter peaks and lighter tails from normal distribution. In our data series used in this study have positive kurtosis values indicate distribution are not normal and follow t- distribution.

4.2 Correlation Matrix

The existence of correlation among different variable of the analysis produce unbiased regression results. This is a common consequence of multicollinearity problem where the standard errors of parameter estimates become very large and the individual effect of the explanatory variables is concealed. This means that additional variables bring no additional information to the analysis. Anderson et al., (2008). Hair et al., (2006) argued that correlation coefficient below 0.9 may not cause serious multicollinearity problem. In addition, Malhotra, (2007) stated that multicollinearity problems exist when the correlation among variables become greater than 0.75. Correlation of each variable with itself gives the value of 1. Otherwise it may range between -1 and 1. The higher values indicate higher correlation the lower value specifies lower correlation. Table 4.2 illustrates the results for this study.

Table 2: Correlation Matrix

Variables	Volatility of Stock Price (VSP)	Oil Prices (OP)	Crude Oil Production (COP)	Real Economic Activity (REA)
Volatility of Stock Price (VSP)	1.000			
Oil Prices (OP)	-0.0888	1.000		
Crude Oil Production (COP)	-0.0147	-0.0676	1.000	
Real Economic Activity (REA)	0.0088	0.1002	-0.3843	1.000

Source; Author Own Computation

4.3 Augmented Dickey-Fuller Test Result (ADF-Unit Root Test)

According to different authors, it is not mandatory to check the order of integration of variables given for ARDL method (see Ibrahim, 2015). The only constraint of ARDL approach is that it cannot be applied in the presence of 2nd order of integrated variable in the model. In that case the value of F-Statistic become invalid (Ibrahim, 2015; Ilyas, Ahmad, Afzal & Mahmood, 2010). We applied Augmented Dickey Fuller (ADF- 1981) and Phillips-Perron (PP-1986) test to investigate the order of integration of the variables used in the model. We employee both the test with trend and intercept because mostly time series data have significant trend and intercept such as the previous work of (Kakar & Khalil, 2011). The result of both the test i.e. ADF and PP test are reported in the table 4.3 which shows that all the variable under consideration are integrated of order I(1) except real economic activity which is stationary at level. None of the variable is integrated of order I(2) and we could proceed to ARDL bound test approach to examine the long run relationship among the projected variables.

Table 3: Unit Root Test Results

	Augmented Dickey Filler (ADF) Test			Phillip Perron (PP) Test		
	At Level	At 1 st Difference	Status	At Level	At 1 st Difference	Status
Volatility of Stock Prices (VSP)	-15.845 (0.0000)	-12.849 (0.0003)	I(0)*	-15.916 (0.0000)	-10.092 (0.0042)	I(0)*
Oil Prices (OP)	-13.035 (0.0000)	-10.812 (0.0000)	I(0)*	-13.210 (0.0000)	-15.808 (0.0038)	I(0)*
Crude Oil Production (COP)	-3.092 (0.110)	-12.932 (0.0000)	I(1)*	-.971 (0.124)	-14.839 (0.0000)	I(0)*
Real Economic Activity (REA)	-2.730 (0.0705)	-10.853 (0.0000)	I(0)*	-3.262 (0.0754)	-10.820 (0.0000)	I(0)**

4.4 Bound Test Result for Non-Linear Co-integration

To examine the long run relationship among stock price volatility, oil supply shocks, oil demand shocks and oil specific demand shocks, we have applied the bound testing approach. The bound test approach involves the F-statistic against the bound critical value as well as sensitive to lag selection (Bahmani-Oskooee and Nasir, 2004; Bahmani-Oskooee and Bhol, 2000). As pointed out by Stock and Watson (2012), it is very difficult to capture important information from the model by including few lag and on the other hand, model become over-fits by including extra lag in the model. To select the optimal lag structure of the model we have utilized AIC and SBC criteria given by Pesaran et al. (2001). The result of both AIC and SBC suggest 2 lag for dependent and independent variable in the model and the result of bound test are given in the Table 4.4 indicates that there is a long run relationship among the projected variables. The calculated value of F-statistic exceeds the upper bound critical value. The distribution of F-statistics depend upon four factors, including the number of explanatory variable included in the model, order of integration of the variable has been included in the model, whether the model incorporate intercept/ trend and finally the sample size (Boutabba, 2014).

Table 4: Bound Test Result

F-Statistic	I(0) Bound (Lower Bound)	I(1) Bound (Upper Bound)	Conclusion
19.4824	3.23	4.35	Co-Integrated

Author`s Computation

4.5 Non-Linear ARDL Estimation Result

The long run as well as short run results are presented in the table 4.5. The entire estimated coefficient is statistically significant and carried a correct sign as it was expected, except oil

specific demand shock which show statistically insignificant. The estimated long run coefficient revealed that volatility of stock prices responded negatively to positive shock of oil supply and statistically significant at 10% level of significant. The relationship implies that discovery of new oil well enhanced oil supply in the market which reduced the volatility of stock market prices. The other possible reason could be the increase in oil supply decrease the monopoly power of the OPEC countries over oil supply which reduces the volatility of stock market prices. While negative oil supply shocks have positive impact on the stock prices volatility in the long run. This result implies that decrease in oil supply to the world market increase the prices of oil. This hike in prices of oil become as a source of high inflation in the domestic market which create economic uncertainty and raises volatility of stock prices. The cumulative multiplier effect of negative shock of oil supply is greater than positive shock of oil supply. The difference in the size of multiplier effect clearly indicates asymmetry in the response of volatility of stock prices to oil supply shocks. This result of the study is consistent with previous finding of the (Kilian 2009; Kilian and Park 2009; Effiong 2014).

On the contrary, both positive and negative shocks of oil demand have positive effect on stock prices volatility in case of Pakistan. The positive shocks of oil demand implies that increase world aggregate demand which shows increase in real economic activities tend to increase stock market volatility. Because, positive aggregate demand shock can be consider as good news for stock market, in the event of positive aggregate demand shock, uncertainty about future cash flows decreases which lead to reduce stock market volatility. One can argue that increase in real economic activity is a sign associated with more stable environment which in turn reduces uncertainty in the stock market prices, (Filis *et al.* 2014). While in case of Pakistan, the result is contrary because positive oil demand shocks increase precautionary demand for oil due to

uncertainties which lead to increase volatility of stock market. (Kilian 2009; Kilian and Park 2009; Effiong 2014). On the other hand, oil demand negative shocks is consider as negative news about world real economic activity which lead to increase stock market volatility, such as those during the Asian crises in 1997 and the credit crunch in 2008 tend to increase stock market volatility. The finding of this is align with, Bloom (2009).

Finally, oil specific demand both positive and negative shocks has statistically insignificant impact on stock price volatility. The primary reason behind the insignificant effect of specific oil demand shock on volatility of stock prices is nature of firm responses to oil changes. Most of the firms nowadays engage in effective hedging strategies which reduce the effect of adverse oil price movements, (see Arouri, 2011) primarily caused by oil specific demand shocks. In short run all the variables have positive impact on the volatility of stock prices except negative shock of oil supply.

The adequacy of dynamic specification has been investigated by various diagnostic tests. We employ Breusch-Godfrey Serial Correlation LM Test, Heteroskedasticity Test, Ramsey RESET test and Jarque-Bera test to investigate normality issue in the residuals, functional form misspecification respectively. The result of all the diagnostic tests are reported in the table 4 and it can be easily concluded from the result that our model pass the entire diagnostic test. The coefficient of determination (R-Square) shows that 57% of the variations in the volatility of stock prices is explained by the explanatory variables which is still above 50% even after taking into consideration the degree of freedom. While adjusted coefficient of determination (adjusted R-Square) still show that 53% variation in the volatility of stock prices is explained by explanatory variables. The coefficient of determination result is confirmed by F-statistics which is 13.22 (0.0000).

Table 5: Nonlinear ARDL estimation results

Variables	Coefficients	T-Statistics	Prob*
Long Run Coefficients			
Constant	46.173554	6.31	0.000
VSP(-1)	-1.270372	-11.63	0.002
OSS ⁺ (-1)	-0.281001	-1.70	0.100
OSS ⁻ (-1)	16.126712	2.05	0.042
ODS ⁺ (-1)	16.75187	2.13	0.034
ODS ⁻ (-1)	6.450799	1.70	0.091
OSDS ⁺ (-1)	-1.424693	1.23	0.222
OSDS ⁻ (-1)	-1.453802	-1.25	0.213
Short run Coefficients			
DVSP	0.122947	1.69	0.093
DOSDS-POS	-1.23630	-1.14	0.257
DOSDS-POS(-1)	-0.08256	-1.14	0.257
DOSDS-NEG	-0.19122	-0.17	0.866
DOSDS-NEG(-1)	0.026044	0.21	0.831
DOSS-NEG	-6.23345	-3.47	0.002
DOSS-NEG(-1)	-5.42020	-2.58	0.001
DOSS-POS	2.650808	2.04	0.030
DOSS-POS(-1)	1.205034	-1.33	0.740
DODS-POS	14.50454	2.24	0.007
DODS-POS(-1)	8.53143	3.03	0.003
DODS-NEG	6.34611	1.96	0.077
DODS-NEG(-1)	2.20584	2.98	0.005
Diagnostic Tests			
R-squared	0.5781		
Adjusted R-squared	0.5343		
S.E of Regression	2725.2		
F-statistics	13.22		
Prob(F-statistics)	0.0000		
A. Serial Correlation	X^2_{auto}	34.49(0.7163)	
B. Heteroscedasticity	X^2_{Het}	0.4422(0.8334)	
C. Functional Form (Remsey Reset)	X^2_{RESET}	1.927(0.1266)	
D. Normality	X^2_{Norm}	0.332(0.6832)	

Source; Author Own Computation

4.6 Wald Test result for Asymmetric Effect

Asymmetric relationship can be observed from the size of the coefficient, however we employed Wald test to investigate the asymmetric relationship among the projected variable. The result of Wald test is reported in the table 4.6, which indicate that oil demand shock and oil supply shock have asymmetric effect on stock price volatility in long run while oil specific demand shock effect are linear in the longer term. On the other hand, the short run result indicates that there is no asymmetric effect except oil demand shock. The result of Wald test is also confirmed by size of long run coefficient which shows that oil supply shocks and oil demand shock coefficient are not same means that there is asymmetry in their effect.

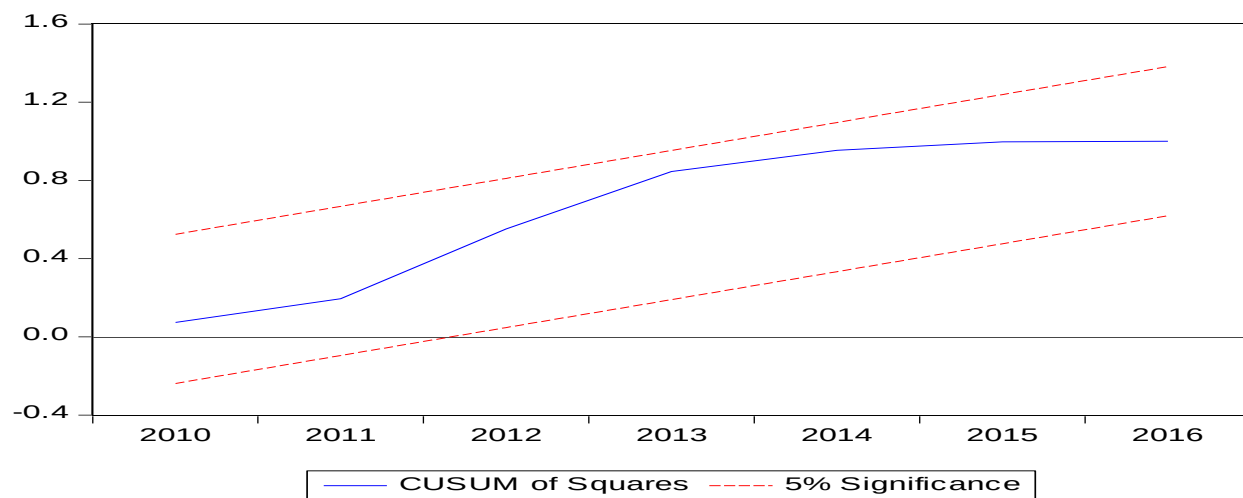
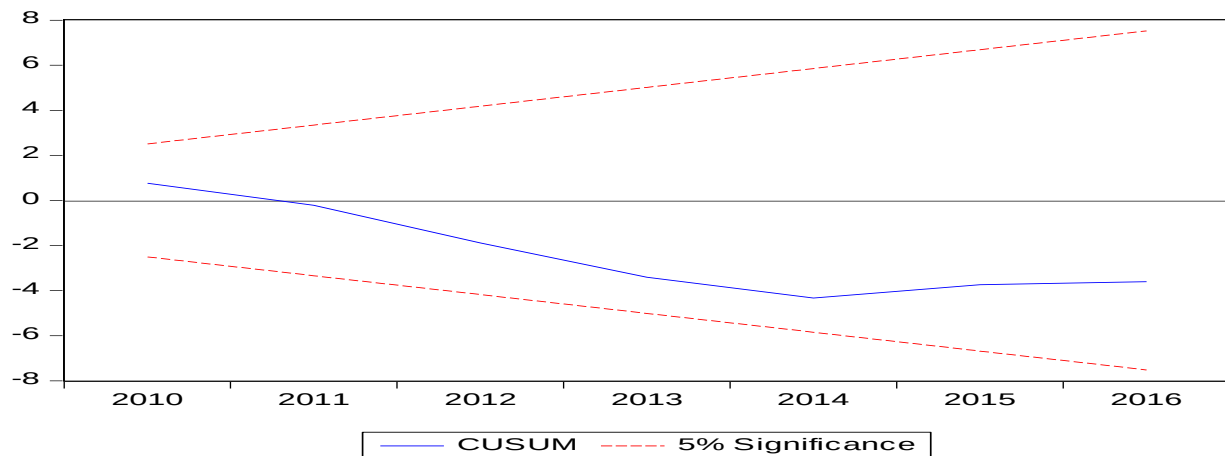
Table 6: Wald Statistics Test Result						
	Long Run Asymmetry			Short Run Asymmetry		
Variable	Wald Value	Wald Prob. Value	Conclusion	Wald Value	Wald Prob. Value	Conclusion
OSS	3.664	0.057	Asymmetric	0.0162	0.899	No asymmetric
ODS	3.938	0.049	Asymmetric	2.998	0.085	Asymmetric
OSDS	0.02677	0.870	No asymmetric	1.219	0.271	No asymmetric

Source; Author Own Computation

4.7 Model Stability Test

The graphical representation of Cusum and Cusum of square test is reported in the figure 3. It is very important to check the strength of parameter stability after the short run and long run coefficient estimation. Brown, Durbin & Evans (1975) and Pesaran & Pesaran (1997) argued and recommended Cusum and Cusum of Square test developed by Brown et al. 1975). The Cusum and Cusum of square statistic repeatedly restore when found any breakpoint. The blue line

represents the Cusum and Cusum of square line while the two red line shows the 5% upper and lower critical bounds. The blue line lies within the upper and lower critical bounds means that estimated parameter is relatively stable.



CHAPTER: 5

CONCLUSION AND POLICY RECOMMENDATION

Economic activities are highly sensitive to the changes in oil prices. Therefore, oil price shocks are worth significant to study. Previously studies have a focus on symmetric effects of oil price shocks however they have drawback that they ignore the dynamics which contribute in asymmetry. This study tries to check out the asymmetric effects of oil price shocks on stock market volatility with the help of newly developed technique Non-Linear ARDL model developed by shin et al. (2014). NARDL have a profound and sound feature which helps to decompose the effect of oil price shocks into positive and negative components to see out the evidence of asymmetry. This study is conducted for the case of Pakistan from 2001 to 2018 monthly data in which three different oil price shocks has been taken including demand, supply and oil specific demand shocks. Furthermore, all three shocks have been decomposed into POS and NEG shocks with the help of NARDL model. Bound test for nonlinear co-integration result indicate that there is long run association among the projected variables. Conclusion of this study can be summarized as, the results of NARDL model shows that except oil specific demand shock all variables have a short run impact on stock market volatility in Pakistan with different lags structure which may supports the claim of short run asymmetry. In long run from, all the variable except oil specific demand shock have a significant impact on stock price volatility with different magnitude which may contribute in asymmetry. This asymmetry impact of projected variables is checked with the help of WALD test. The result shows that in long run, oil supply shock, oil demand shock has an asymmetric effect on volatility of stock market in Pakistan whereas, oil specific demand shock test statistics indicate the absence of asymmetric effect. While in short run only oil demand show an asymmetric impacts and rest of the two variables namely oil supply

shock and oil specific demand shocks results indicate that there is no asymmetric effect on volatility of stock prices. Hence, they have a different policy implication.

5.1 Policy Recommendation

This study suggests the policies on the basis of underlying theory and their extracted results for the impact of oil price shocks including demand shocks, supply shocks and oil specific demand shocks to the volatility of stock market for Pakistan. The study has checked the asymmetric effects of these shocks on volatility and found that each shock reacts differently to the stock market volatility depending on the nature and source of shocks. This study suggests the investor that must take into consideration the asymmetry of these shocks before going to invest in market as it has a significant importance in market. Manager should also consider the asymmetric nature while taking decisions regarding business as its activity mainly deals with the transportation which will be affected by these oil price shocks. Producers have most concern to all types of shocks as their cost of production varies due to oil price shocks if these shocks have asymmetric nature then it will hurt the cost of production and leads to decrease in firms revenue therefore producers must take this asymmetry into account while expanding their business and production. Other than these consumers should also remain conscious about the changes in oil price as it hurts their purchasing power due to reliance of industries on oil which may boost the price of products due to changes in oil price shocks. These asymmetric shocks may be a very important and good source for petroleum industry to enhance their profit as to take them before setting the price.

REFERENCES

- Abken, P. A, and S Nandi. 1996. "Options and volatility." *Economic Review* 81: (3-6).
- Adeniji, S. O. 2014. "An Appraisal of Stock Prices Volatility in the Nigerian Stock Market in an Era of Democracy." *Unpublished MSc. Thesis, Department of Economics, University of Lagos*.
- Al-hajj, Ekhlas, Usama Al-Mulali, and Sakiru Adebola Solarin. 2018. "Oil price shocks and stock returns nexus for Malaysia: Fresh evidence from nonlinear ARDL test." *ELSVIER* 624-637.
- Ansar, I, and N Asghar. (2013). "The impact of oil prices on stock exchange and CPI in Pakistan." *IOSR Journal of Business and Management (IOSR-JBM)* E-ISSN: 32-36.
- Arouri, M. E. H, and C Rault. 2012. "Oil prices and stock markets in GCC countries: empirical evidence from panel analysis." *International Journal of Finance & Economics* 17(3): 242-253.
- Arshad, R. A. B. I. A, and A. D. N. A. N Bashir. (2015). "Impact of oil and gas prices on stock returns: Evidence from Pakistan's energy intensive industries." *International Review of Social Sciences* 3(4): 156-168.
- Bastianin, A, Conti F, and M Manera. 2016. "The impacts of oil price shocks on stock market volatility: Evidence from the G7 countries." *Energy Policy*, 98, 160-169 98: 160-169.
- Broadstock, David, and George Filis. 2014. " Oil price shocks and stock market returns: New evidence from the United States and China." *International Financial Market, Institutions and Money*.

- Degiannakis, Stavros, George Filis, and Renatas Kizys. 2014. "The effects of oil price shocks on stock market volatility: Evidence from European data." *The Energy Journal*.
- Fang, Chung-Rou, and Shih-Yi You. 2013. "The Impact of Oil Price Shocks on the Large Emerging Countries' Stock Prices: Evidence from China, India and Russia." *International Review of Economics and Finance* 1-28.
- Filis, G. 2010. "Macro economy, stock market and oil prices: do meaningful relationships exist among their cyclical fluctuations?." *Energy Economics* 32(4): 877-886.
- Hamilton, J. 2009. "Understanding Crude Oil Prices." *The Energy Journal* 30 (2) 30: 2-9.
- Hu, Chunyan, Xinheng Liu, Bin Pan, Bin Chen, and Xiaohua Xia. 2017. "Asymmetric Impact of Oil Price Shock on Stock Market in China: A Combination Analysis Based on SVAR Model and NARDL Model." *Emerging Markets Finance and Trade*.
- Jones, C. M, and G Kaul. 1996. "Oil and the stock markets. The journal of Finance, 51(2), 463-491." *The journal of Finance* 51(2): 463-491.
- Kang, Wensheng, Ronald A. Ratti, and Kyung Hwan Yoon. 2015. "The impact of oil price shocks on the stock market return and volatility relationship." *International Financial Market. Institution and Money* 34: 41-54.
- Kilian, L. 2009. "Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market." *American Economic Review* 99(3): 1053-69.
- Kilian, L. 2008. "The economic effects of energy price shocks." *Journal of Economic Literature* 46(4): 871-909.

- Marashdeh, Hazem, and Akhsyim Afandi. 2017. "Oil Price Shocks and Stock Market Returns in the Three Largest Oil-producing Countries." *International Journal of Energy Economics and Policy* 7 (5): 312-322.
- Masih, R, S Peters, and L De Mello. 2011. "Oil price volatility and stock price fluctuations in an emerging market: evidence from South Korea." *Energy Economics* 33(5): 975-986.
- Mendoza, O, and D Vera. (2010). "The asymmetric effects of oil shocks on an oil-exporting economy." *Cuadernos de economía* 47(135): 3-13.
- Miller, J. I, and R. A Ratti. 2009. "Crude oil and stock markets: Stability, instability, and bubbles." *Energy Economics* 31(4): 559-568.
- Mollah, S. A. 2009. "Stock Return & Volatility in the Emerging Stock Market of Bangladesh." *Journal of the Academy of Business & Economics*, 43(2): 29-78.
- Naurin, A., & Qayyum, A. (2016). "Impact of Oil Price and Its Volatility on CPI of Pakistan: Bivariate EGARCH Model."
- Oluseyi, A. S. 2015. "An empirical investigation of the relationship between stock market prices volatility and macroeconomic variables' volatility in Nigeria." *European Journal of Academic Essays* 2(11): 1-12.
- Ono, S. (2011). "Oil price shocks and stock markets in BRICs." *The European Journal of Comparative Economics* 8(1): 29-45.
- Park, J, and R. A Ratti. 2008. "Oil price shocks and stock markets in the US and 13 European countries." *Energy economics* 30(5): 2587-2608.

- Ratti, R. A, and M. Z Hasan. 2013. "Oil price shocks and volatility in Australian stock returns." *Economic Record* 89: 67-83.
- Sadorsky, P. 1999. "Oil price shocks and stock market activity." *Energy economics* 21(5): 449-469.
- Siddiqui, Rehana. 2004. "Energy and Economic Growth in Pakistan." *Pakistan Development Review* 175-200.
- Syzdykova, Aziza. 2018. "The Relationship between the Oil Price Shocks and the Stock Markets: The Example of Commonwealth of Independent States Countries." *International Journal of Energy Economics and Policy* 6 (8): 161-166.
- Bahmani-Oskooee, M, and A. B. M Nasir. (2004). "ARDL approach to test the productivity bias hypothesis ." *Review of development Economics* 8(3): 483-488.
- Bjørnland, H. C. 2009. "Oil price shocks and stock market booms in an oil exporting country." *.Scottish Journal of Political Economy* 56(2): 232-254.
- Bloom, N. (2009). "The impact of uncertainty shocks." *econometrica* 77(3): 623-685.
- Boutabba, M. A. (2014). "The impact of financial development, income, energy and trade on carbon emissions: Evidence from the Indian economy." *Economic Modelling* 40: 33-41.
- Brown, R. L, J Durbin, and J. M Evans. (1975). "Techniques for testing the constancy of regression relationships over time." *Journal of the Royal Statistical Society: .Series B (Methodological)* 37(2): 149-163.

- Engle, R. (2002). "New frontiers for ARCH models." *Journal of Applied Econometrics* 17(5): 425-446.
- Huang, R. W Masulis, and H. R Stoll. (1996). "Energy shocks and financial markets." *Journal of Futures Markets: Futures, Options, and Other Derivative Products* 16(1): 1-27.
- Ibrahim, M. H. (2015). ".Oil and food prices in Malaysia: a nonlinear ARDL analysis." *Agricultural and Food Economics* 3(1): 2.
- Ilyas, M, H. K Ahmad, M Afzal, and T Mahmood. (2010). " Determinants of manufacturing value added in Pakistan: An application of bounds testing approach to cointegration." *Pakistan Economic and Social Review* 209-223.
- Katrakilidis, C, and E Trachanas. (2012). "What drives housing price dynamics in Greece: New evidence from asymmetric ARDL cointegration." *Economic Modelling* 29(4): 1064-1069.
- Khalil, S, and M. K Kakar. (2011). "Agricultural use of untreated urban wastewater in Pakistan." *Asian Journal of Agriculture and Rural Development* 1(393-2016-23907): 21-26.
- Kolodzej, M, and R. K Kaufmann. (2014). "Oil demand shocks reconsidered: a cointegrated vector autoregression." *Energy Economics* 41: 33-40.
- Malhotra, N, and D Birks. (2007). " Marketing Research: an applied approach." *3rd European Edition. Pearson education.*
- Pesaran, M, Y Shin, and R. J Smith. (2001). "Bounds testing approaches to the analysis of level relationships." *Journal of applied econometrics* 16(3): 289-326.

- Shin, Y, B Yu, and M Greenwood-Nimmo. (2014). "Modelling asymmetric cointegration and dynamic multipliers in a nonlinear ARDL framework." *In Festschrift in honor of Peter Schmidt* 281-314.
- Stock, J. H, and M. W Watson. (2012). "Generalized shrinkage methods for forecasting using many predictors." *Journal of Business & Economic Statistics* 30(4): 481-493.
- Wei, Y, X Liu, H Song, and P Romilly. (2001). "Endogenous innovation growth theory and regional income convergence in China." *The Journal of the Development Studies Association* 13(2): 153-168.